Equivariant Cohomology

Reference is Goresky

Plan
1) Defn, examples
2) Localization thm, application
3) Examples: flag, Gr.

time permitting
4) Kirwan dual eq. coh., examples, higher order coh. op
5) Kirwan duality.

Convention: K compact Lie group, or sometimes its complexification

$K = (S^1)^n \cong (\mathbb{C}^*)^n$

Example: 1) $H^\cdot_K(K) = \mathbb{C}$ $\quad$ $H^\cdot(\mathbb{C}^n - 0/\mathbb{C}^*)$

2) $T = S^1$. $H^\cdot_T(pt) = H^\cdot(BS^1) = H^\cdot(\mathbb{C}P^\infty)$. $\quad (ES^1 = S^\infty \simeq \mathbb{C}^\infty - 0)$

$= \mathbb{C}[\beta]$ $\quad \deg(\beta) = 2$.

$T = (S^1)^n$, $H^\cdot_T(pt) = S\mathfrak{t}^*$.

Equivariant Chains: "more geometric" (slightly) $\dim(K) = k$

Def^n K-equivariant i-chain:

"reasonable space" → space C with free K-action, $\dim i+k$

K-eq map $C \to X$.

Prop Two definitions equivalent, if smooth chain complex.

Example: $T = S^1$, $X = pt$. Then $\beta^k \in H^\cdot_T(pt)$ is

$S^1 \circlearrowright S^{2n+1} \longrightarrow pt$. $\quad$ (from $\mathbb{C}^* \circlearrowright \mathbb{C}^n - 0$)

1) Bad Def^n $H^\cdot_K(X) = H^\cdot(X/K)$ $\quad$ ($K \circlearrowright X$)

Why? 1) Lose the group K e.g. $X = pt$ vs $X = K$.

2) If action is not locally free, X/K bad.

Good Def^n "Borel construction".

Define $H^\cdot_K(X) = H^\cdot(X \times_K EK) = H^\cdot((X \times EK)/K)$

diagonal action

[universal bundle EK (contractible, $K \circlearrowright EK$ free)]

Note: $\widetilde{pt} = EK/K = BK$ classifying space.

Remark If $K \circlearrowright X$ is free, $H^\cdot_K(X) = H^\cdot(X/K)$.

Example: $S^1 \circlearrowright S^3$ trivial. Then $H^\cdot_{S^1}(S^3) = H^\cdot(S^3) \otimes k[\beta]$.

$S^1 \circlearrowright S^3$ Hopf; free action, so $H^\cdot_{S^1}(S^3) = H^\cdot(S^2) = k[\beta]/\beta^2$.

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2) Localization Theorem

Thm Suppose that $H^\cdot_T(X)$ is free / $H^\cdot_T(pt)$. Let F = fixed pt set, X_1 = set of $\leq 1-1$ orbits, all finite. Then,

$$0 \to H^\cdot_K(X) \to H^\cdot_K(F) \to H^\cdot_K(X_1, F)$$

exact.

Prop If X is a smooth projective complex variety, then $H^\cdot_T(X)$ is free.

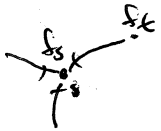
Example: $S^1 \circlearrowright \mathbb{C}P^1$.

How to use:

1) $H_T^\cdot(X) \xhookrightarrow{\text{sub}} H_T^\cdot(F) = \bigoplus_{x\in F} St^*$. Draw node $\forall x \in F$.

2) Each 1-d orbit $\simeq \mathbb{C}^*$; two fixed points in closure. Draw edge.

3) $H_T^\cdot(F) \to H_T^\cdot(X_1, F)$ surjectivity.



Locally, $F = \{s,t\}$, orbit $O \cong T/\mathrm{Stab}(O)$.

$k[t] \otimes \{s,t\} \longrightarrow k[\mathfrak{s}] = H_T^\cdot(O)$ where $\mathfrak{s}$ = Lie algebra of column stab.

$f_s s + f_t t \longmapsto (f_s - f_t)|_{\mathfrak{s}}$.

Want: $f_s - f_t$ zero on $\mathfrak{s}$, i.e. $f_s - f_t \in S\mathfrak{s}^\perp$.

Generated by ~~this~~ single vector $t \in t^*$. Local edge

3) Example: Flag variety SL_3/B.

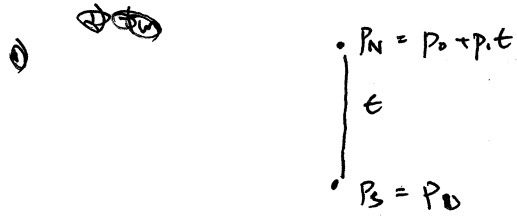
$T = (S^1)^2$ generators: $\begin{pmatrix} x & & \\ & x & \\ & & x^{-2} \end{pmatrix}$, $\begin{pmatrix} y^{-2} & & \\ & x & \\ & & x \end{pmatrix}$ $\in T$ $\rightsquigarrow$ basis $t_1\begin{pmatrix} a & & \\ & b & \\ & & c \end{pmatrix} = a+b-2c$, $t_2\begin{pmatrix} a & & \\ & b & \\ & & c \end{pmatrix} = b-c-2a$ $\in t^*$.

Fixed points: permutation matrices.

Orbits: $\begin{pmatrix} 1 & & \\ a & 1 & \\ & & 1 \end{pmatrix}$ not: $\begin{pmatrix} 1 & a & \\ & 1 & \\ & & 1 \end{pmatrix}$ (w/c Borel)

$a \to \infty$: $\begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}$

Example: $S^1 \circlearrowright \mathbb{CP}^1$ by rotation



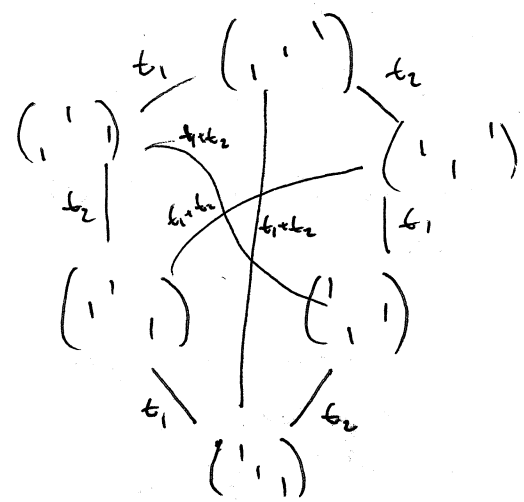
$p_N - p_S = p_1 t$

So, basis: $\begin{pmatrix} p_N \\ p_S \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} t \\ 0 \end{pmatrix}$.

$[S^2]$ $[pt]$ (Poincaré duality.)

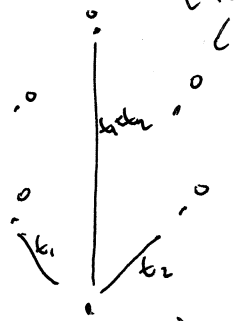
So, $H_{S^1}^\cdot(S^2) = H^\cdot(S^2) \otimes k[\beta]$.

Schubert calc: $\overrightarrow{BwB}$.



Schubert cells

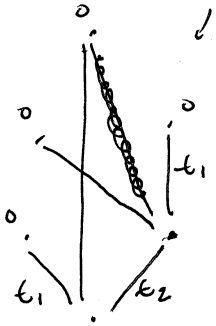
$[\sigma_{123}]$



$t_1t_2(t_1+t_2)$

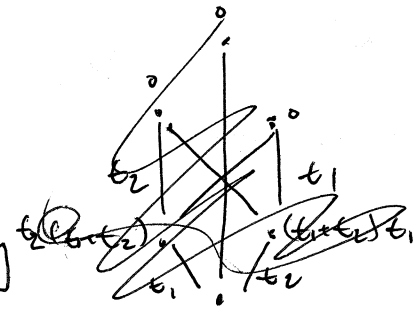
(bottom dim homology
top dim cohomology)

$[\sigma_{132}]$

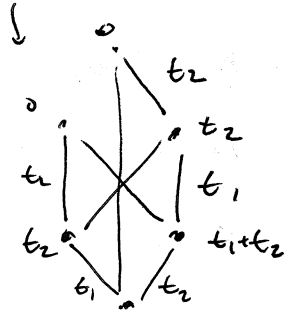


$t_1(t_1+t_2)$

$t_1(t_1+t_2)$

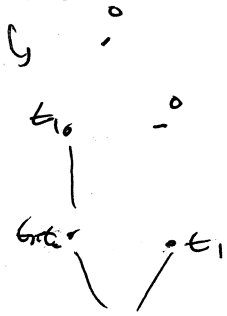


$[\sigma_{231}]$



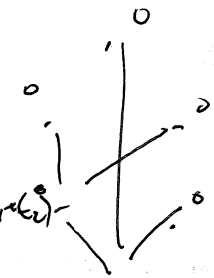
t_1+t_2

$[\sigma_{312}]$



t_1+t_2

$[\sigma_{213}]$



$t_2(t_1+t_2)$

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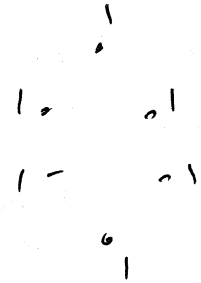
$[\sigma_{321}]$



Grassmannian: top dim: $\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \cdot & \cdot \\ & & \cdot & \cdot \end{pmatrix} \rightsquigarrow$ 0011 Φ = no stars

bottom: $\begin{pmatrix} & & 1 & \\ & & & 1 \end{pmatrix} \rightsquigarrow$ 1100 X_{12} "a" in any cal w/v 1

$T = \mathbb{C}(S^1)^4/S^1$.

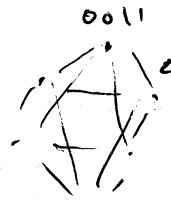
Prop 1. ~~Schubert~~ Checks:

$[\sigma_{132}] \cdot [\sigma_{213}] =$

$= (t_1+t_2)[\sigma_{123}]$

$t_i = 0 \rightsquigarrow 0$

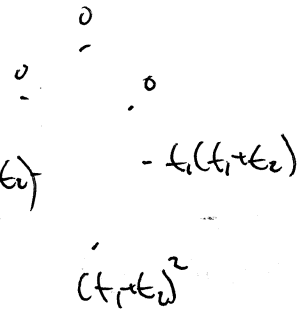
$t_1t_2(t_1+t_2)^2$



$t_i - t_j$ swaps i, j.

$(y_3-y_1)(y_4-y_1)(y_3-y_2)(y_4-y_2)$

$[\sigma_{231}] \cdot [\sigma_{312}] =$



$= [\sigma_{132}] + [\sigma_{213}]$

$t_2(t_1+t_2)$

$-t_1(t_1+t_2)$

$(t_1+t_2)^2$

etc.....

left as exercise?

4) Koszul dual def'n: even more geometric! pontrjagin action on chains

$C_\cdot(K) \otimes C_\cdot(X) \to C_\cdot(X)$ "sweep" (zero if dim is wrong)

In general, $C_\cdot(K)$ coalgebra. $C_\cdot(K)$ is Hopf algebra.

Example: $T = (S^1)^n$. $H_\cdot(T) = \Lambda t$.

Example: $S^1 \curvearrowright S^3$ trivially: $H_\cdot(S^1) \curvearrowright H_\cdot(S^3)$ trivially.

$S^1 \curvearrowright S^3$ Hopf. $H_\cdot(S^1) \curvearrowright H_\cdot(S^3)$ trivially!!!

Higher cohomology actions: $\mu_3(\overset{H(S^1)}{[S^1]}, \overset{H(S^1)}{[S^1]}, \overset{H(S^3)}{[pt]}) =$ sweep pt to $S^1 \subseteq S^3$
kill with cohomology
sweep cohomology
$\rightsquigarrow [S^3]$!

← this is exactly obstruction for $C^\cdot(X) \simeq H^\cdot(X)$

Higher cohomology actions detect failure of $H_\cdot(S^3)$ to be $H_\cdot(S^1)$ qio $C_\cdot(S^3)$ over $H_\cdot(S^1)$. (A_∞-model).

4) Koszul duality

$D^b(St^a) \rightleftarrows D^b(\Lambda t)$ equivalence

free (formal) $\longleftrightarrow$ trivial (formal) ← so we can talk about coh.

So: free St^a coalgebra (formal) = trivial Λt coalgebra (formal) = basis of K-closed chains + higher coh operations vanish (A_∞-module)

K-closed w/ basis

For compact (projective) $\mathbb{C}$-alg variety, this is always true, by necessity of degrees.

i.e. μ_n degree $2-n$. $\underbrace{[S^1] * \cdots [S^1]}_{\text{odd}} * \underset{\text{even}}{[S^3]}$ is degree $(n+1 + \text{even}) + 2 - n = 3 - 2n + \text{even} = \underline{\text{odd}}$!

~~Idea of pf of Ka thm: Supp A on Spec $\mathbb{C}[t]$~~

~~Say $\mathcal{O}$ is coh in X, with stab. S.~~

~~Then $K = S \times \mathbb{P}$. $S \times \mathbb{P}^\times \subseteq St^a$ should act by tensoring.~~

~~So, if $H^i_T(X)$ torsion-free,~~

Note: formal $\not\longleftrightarrow$ formal!